21-242 Recitation 12

RREF — Q4. Example with (T A) (T (A))

Course-level solution, with a later-course alternative when useful

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Prompt

Find an example of A with $\mathrm{im}(T_A) \neq \mathrm{im}(T_{\mathrm{RREF}(A)})$.

Course-level solution

Row operations change column space in general. Take

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\mathrm{im}(T_A) = \mathrm{Span}\{(1,1)^T\}$ while $\mathrm{im}(T_{\mathrm{RREF}(A)}) = \mathrm{Span}\{(1,0)^T\}$, which are unequal.